

Scalability & Future Plan

Date	11 July 2026
Team ID	
Project Name	EduGenie Google Gemini Powered Learning Assistant
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High/Medium/Low)
1	SQLite database used for development is not built for concurrent high-volume access	Limits number of simultaneous users the system can reliably support	High
2	No caching layer for repeated/common Gemini API queries	Increases response time and API cost for frequently asked questions	Medium
3	Currently supports English-language queries only	Excludes non-English speaking students from using the platform	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single FastAPI instance, handles limited concurrent users	Deploy behind a load balancer with multiple containerized FastAPI instances using Docker
2	Data Storage	SQLite local file-based database	Migrate to PostgreSQL with proper indexing for production-scale concurrent access
3	Performance	Direct Gemini API call for every query, no caching	Implement Redis caching for frequently asked questions to reduce latency and API cost
4	Security	Basic .env-based API key protection	Add rate limiting, JWT-based session auth, and API request logging for monitoring

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Migrate database from SQLite to PostgreSQL for production readiness	1-2 months	Improved reliability and support for concurrent users
Phase 2	Add multilingual support (Hindi, Spanish) for wider accessibility	3-4 months	Expands user base to non-English speaking learners
Phase 3	Introduce mobile app version (Android/iOS) using Flutter	5-6 months	Increases accessibility and daily active usage
Phase 4	Add voice-based Q&A and gamified quiz leaderboards	7-9 months	Boosts engagement and makes learning more interactive